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Education

The race for AI skills as an obstacle course: Institutional challenges and low threshold suggestions

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ABSTRACT

Overall, the global labor market shows a clear surge of demand for AI-related skills, pressuring education systems and institutions across the globe, in particular those in higher education, to adapt policies and curricula and address the current skills gap. This paper examines the delaying factors which higher education institutions are facing in heeding a swift call for action, including problems related to the current AI skills frameworks, the heterogeneity of student populations, the lack of institutional policies as well as the slow process of translating new knowledge into curricula. The paper concludes with some suggestions for low threshold actions, which would help prevent the skills gap from becoming even wider in the near future as well as pave the way for more holistic solutions.

1. Introduction: the rise of and race for AI skills

With the rise of generative artificial intelligence (GenAI) in public discourse and consciousness, since ChatGPT's release for public use on November 30, 2022, expectations have been hyperboling. In 2023, which McKinsey labelled the "*break out year*" for GenAI, about one third of the 2395 business leaders from various industries and regions which the company had interviewed on their use and views of generative AI, reported that their organizations were using generative AI regularly in at least one business function. One year later that share had climbed to more than 60 % (McKinsey, 2023, 2024). The more recent survey also shows that the use of AI tools across different types of business is not only becoming more frequent, but also more varied – yet at the same time, incorrect use was also becoming more of a risk (McKinsey, 2024). It is also worth noting that the majority of explicitly AI-related skills sought in current job advertisements appear to come from industries and organizations dabbling in AI development and implementation, as can be deduced from a recent OECD comparative study on emerging trends across fourteen different countries (Borgonovi et al., 2023). By contrast, predictions for future demand, probably related to the integration of generative AI applications in everyday working routines across professions and business fields, place a slightly greater emphasis on soft skills (Bukartaite and Hooper, 2023).

Overall, the global labor market shows a clear surge of demand for AI-related skills – although it is important to note that such an increase has already been observed for the past ten to fifteen years, albeit not at the same level (Alekseeva et al., 2021). However, the call for skills development on a large scale is not only economically motivated: The EU AI Act, issued in 2024 and intended to protect EU citizens and beyond from the abuse and misuse of AI, requires both providers and users of AI applications to ensure that actors on both sides have a *sufficient* level of AI literacy (Regulation, 2024/1689).

This rising demand poses quite a challenge for educational "suppliers": As Jaiswal et al. (2024) show, there is a certain skills gap between what businesses need and what higher education institutions (can) provide, not least because the current higher education provision only covers a fraction of disciplines. Accordingly, education systems and institutions across the globe, in particular those in higher education, are facing increased pressure to adapt policies and curricula, as the solution for overcoming a skills gap in industry appears to be *lying in education* (Chrisinger, 2019). Even though, as Holmes and Tuomi (2022) put it, part of the pressure to deliver is linked to a misunderstanding of current technical possibilities and what education can and cannot achieve, and a common understanding of what the concept of *skills gap* entails, is actually still lacking (Rikala et al., 2024), scholars and policy makers alike are calling for implementing AI across curricula (Southworth et al., 2023) and for AI educational policies (Chan, 2023), all of them linked to AI literacy frameworks. As the authors of the UNESCO AI competency

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framework for students put it: "Education, as a public sector, cannot be reduced to a testing ground for the passive adoption of AI." (UNESCO, 2024). De Witt et al. (2020) regard an accepted AI skill or competence framework as an indispensable condition for moving beyond such a passive approach and Raley and Rhee (2023) argue that higher education institutions have the responsibility to support staff and students in building AI literacy skills.

The underlying megatrends of working with big/rich data and digitalization are not least changing the project management profession profoundly, leading to similar predictions about skills that are becoming more important, but to some degree still lacking (Janssen et al., 2017; Liu et al., 2024). In their systematic literature review on the transformational power of AI in project management, Nenni et al. (2024) come to the conclusion that AI in project management is predominantly applied in sectors related to construction and IT, with a strong focus on machine learning supported project risk management. Badhon et al. (2025) consequently advocate a multi-module framework to enable a more effective project risk management.

Our paper takes its cue from the overall "call for action" (for the project management context, also cf. Müller et al., 2024) yet shifts the attention to various factors that make an overly speedy and systematic tackling of the skills gap between industry demands and higher education provision unlikely for the near future. Our main research question thereby asks about the conceptual and structural challenges higher education is facing in making students *AI-literate* on a larger scale – and how at least some of these challenges could be overcome by introducing low threshold solutions.

As a starting point, we present different AI literacy/skills frameworks as currently proposed by scholars and policy makers and analyze their key similarities and differences – and the implementation challenges that are built into these kinds of frameworks. In a second part, we will explore challenges that lie outside of the frameworks themselves, but are still endogenous to higher education, such as the great variance of skills and knowledge gaps between and among educators and learners; and the delaying factors related to how new knowledge gets operationalized and "codified" in higher education, e.g. by ways of textbooks and curricula. We deliberately do not tackle other relationships between education and AI, such as AI as a support for teaching delivery, including assessment or AI as infrastructure for educational operations.

2. AI literacy and skills frameworks and the challenges lying therein

The number of AI literacy frameworks and suggestions is quickly growing. Ehlers et al. (2023) have examined more than ten competence/literary approaches which have been published since 2020 – and this is not including research on related areas such as ICT or 21st century skills. For our paper, we have taken a two-fold approach. In a first step, we conducted a literature search via SCOPUS, using search the search terms "'AI skills", "AI comperences" and "AI literacy" (AI abbreviated and fully written). After excluding publications published in another language than English or not yet being available as a final text, 542 texts remained in the corpus. These were further scrutinized via their key words and abstracts, looking for publications on/about AI skills frameworks/AI literacy in higher education. In a next step, we excluded all papers that focused on specific application contexts (e.g. AI skills frameworks for higher education librarians) and frameworks targeting secondary education/K12 cycle or primary education. During the search it became quickly clear that this strategy on its own would not be sufficient, as the turnout proved to be quite meagre, most likely for the fact that academic research on AI skills is still scarce, and respective proposals can for the time being rather found in so called "grey literature" – a hypothesis partly confirmed by a parallel search using the same search terms via Google Scholar. Therefore, we decided to complement our Scopus search with a snowballing/citation chaining approach. Ultimately we ended up with about two dozens relevant framework related publications – yet in cases where the same authors or authoring institution had issued several over a short period of time (such as in the case of UNESCO) which do not differ much, they were analysed as one entity.

Several suggestions for AI-proof curricula and competence frameworks precede the launch of ChatGPT by years, though with a twist: In a literature review spanning 20 years (from 2000 to 2020), Ng et al. (2023) do not only point out how the term *AI Literacy* only surfaced less than ten years ago but show how AI related teaching was mostly limited to computer science classes until 2021, with basic programming skills being a prerequisite for any learner. They regard these pioneering skills development approaches an important basis for further development, but not ready for scaffolding, i.e. able to reach more students with a more heterogeneous background.

In parts, such a shift towards a broader population of learners and a more generic skill set has already started, almost a decade ago. Aoun (2017), for instance, describes three "new literacies" that higher education institutions should foster: Technological literacy, data literacy – and human literacy, i.e. abilities to communicate and engage with other human beings in a meaningful way. Even though artificial intelligence is not name-checked in any of these literacies, it provides the background against which Aoun develops his educational model. Pedagogically, his focus lies on experiential learning. This example also already shows where AI skills frameworks differ from most digital skills and ICT skill frameworks, i.e. in putting equal emphasis on understanding social contexts and phenomena (Farrelly and Baker, 2023). A variant of human literacy is also included in the UNESCO AI competency framework for students (UNESCO, 2024), which outlines twelve competencies organized across four dimensions: A human-centred mindset, ethics of AI, AI techniques and applications, and AI system design. These dimensions are also mirrored in Ehlers' et al.'s AIComp model (2023), which groups the skills differently, but again includes self-/social competence aspects, system thinking, digital literacy and ethics, even though putting the arguably strongest emphasis on innovation and creativity. Ethics is becoming an increasingly important dimension in its own right: As Ng et al. (2023) argue, with AI (potentially) influencing almost every sphere of life, it is pivotal to also educate learners about the ethics behind AI. Floridi et al. (2021) focus their entire framework on ethics, adding a new principle (explicability) to four more traditional ones (beneficence, non-maleficence, autonomy and justice).

Comparing the different takes on AI skills, we observe considerable overlap between the different frameworks, at least on a generic level (such as a focus on social and communicative skills as well as on ethics and critical thinking) but also a great variety when it comes to specifics (different degrees to which ICT skills, but also creativity and innovation are emphasized). From an implementation perspective, we see the following challenges arising from the variance between the various frameworks and the way they are built:

First of all, we acknowledge an emerging consensus that various skills and competence frameworks are not yet sufficiently researched and there is a need to formulate and disseminate respective competence frameworks (Ehlers et al., 2023). In other words, the discourse is still in its infancy, which will impede any early attempt to reach convergence or consensus. Attempts such as the Generative AI Literacy Test proposed by Jin et al. (2024) could, in principle, accelerate the development, but in its current form covers only a fraction of the skills suggested by the different framework proponents. In addition, it is highly likely that the frameworks will be (and partially already are) adapted to specific disciplines and communities of practice (Farrelly and Baker, 2023). And last, but not least, the majority of AI literacy frameworks focuses on the learners, defining what students (or in broader sense people) need to thrive in today's economy and society. Far less attention is paid to the educators and the skills they require to support and empower the learners (Ng et al., 2023), even though they are admittedly the primary target audience for the various student level frameworks and will need to develop a specific set of competencies of their own in order to use AI to its full potential (Kim, 2024).

At the same time, we can also see opportunities, such as certain elements that are shared across the different frameworks, which can to some degree be interpreted as areas most likely to converge. This includes most of all, an emphasis on understanding the human mind and human interaction as much as the technological and algorithmic foundations of AI. What most frameworks also have in common is the call for a critical thinking approach towards AI. The specific terms might differ, but there is a clear overlap with what Dwyer et al. (2014) call a metacognitive process, "(…) consisting of a number of sub-skills (e.g. analysis, evaluation and inference) that, when used appropriately, increases the chances of producing a logical conclusion to an argument or solution to a problem." (Dwyer et al., 2014). The OECD (2005) - even though using the term "reflectiveness" for a similar bundle of skills rather than "critical thinking" – also regards it as a "second order" key competence of higher importance than most other skills. This also mirrors at least partly the discourse on core skills within project management, where transversal skills like problem solving, creativity, and conceptual thinking, are considered important by researchers and practitioners alike (Moradi et al., 2020).

We will come back to framework-derived challenges – as well as the potential lying in the opportunities - in our section on potential steps forward yet first want to explore further factors impeding a swift implementation of AI frameworks, which we regard as endemic to higher education.

3. Structural challenges for higher education institutions in tackling AI skills

In this section, we examine the challenges that higher education institutions are facing in tackling the skill gap mentioned in the introductory part systematically. We structure our arguments along a very basic logic. Every skills development process involves at least three basic components: "Educators/teachers", "students/learners" - and curricula in the broad sense of an ensemble of educational processes, most importantly comprising a *plan for the experiences* that learners will encounter, as well as the design for the *actual experiences* they go through (cf. Remillard and Heck, 2014). This also mirrors our interpretation of the rather sparse existing literature of teachers' and learners' AI proficiency and curricular or pedagogical implementation of AI literacy frameworks: while conceptualisations of AI related skills frameworks increase, research on their implementation still seems in its infancy. Based on the research we identified through our search the first issue is grounded in the great variance of learners and learner populations. The second issue comprises obstacles that might hold back teachers in speedily responding to the skill gap in their teaching. And the third issue touches upon the traditional, but also time-consuming way of "legitimising" and crystallising knowledge by the means of formal curricula.

3.1. Issue one: the differences between learners and their attitudes towards AI

The discourse on AI skills (as any discourse on graduate skills and learning outcomes) typically generalises imparting those skills in a rather simplifying way: All learners need to be supported in developing the same set of skills at the same level. It is important to be aware, though, that learners and their attitudes towards and experiences with GenAI are far less homogeneous than assumed. Published research in peer reviewed journals of some renown is still scarce when it comes to a systematic assessment of students' adoption of AI, but there are several indications that the variance between and within different populations of learners is considerable, with a great number of variables at play, such as gender, age, disciplinary focus or cultural and geographic background.

In an institutional survey with almost 5000 students from different fields, Balabdaoui et al. (2024) for example, observe a tendency that female students are more skeptical towards AI than men and that there is also a disciplinary factor at work, with engineering and biological sciences being more accepting of AI than non-engineering sciences. Strzelecki (2024) shows that to some degree acceptance and use of GenAI differ depending on the degree program level (undergraduate versus graduate). Chan and Lee (2023) go even further in exploring age as, a factor: In investigating potential generational differences, they find that Gen Z students – having grown up with technology and the internet – are more likely to embrace new technological developments than previous generations. Differences in attitudes and perceived skills regarding AI, seem to be in general important influences (Delcker et al., 2024).

Farrelly and Baker (2023) make a case for international students (from an Anglo-Saxon perspective), for whom English likely is not the first language, and who might thus be disadvantaged regarding effective usage and more prone to be branded as "cheaters". Differences in geographical location appears, in general, to be a decisive factor: Following Pham et al. (2024) and their scrutiny of a recent Eurobarometer survey, reveals considerable differences in AI optimism across different European countries – but also between urban and rural communities. Students' national backgrounds also play a key role in the study by Mohamed et al. (2025), comparing student experiences from institutions in Saudi Arabia, Egypt, Spain and Poland.

Other factors that – inferring from previous research on ICT-related skills and research on AI adoption outside of higher education – will likely have an impact on how quickly and well students might be adapting to AI, are socio-economic backgrounds (Bozionelos, 2004), social pressure and norms (Kelly et al., 2023b), and, not least the economic cost of licenses and IT hardware. The research cited only touches on differences between populations of learners – it stands to reason that in practice various variables will interact and reinforce each other, and that individual differences within populations will even outweigh those between populations. In a nutshell, learners show an enormous amount of diversity regarding their knowledge about and inclination to use AI.

Our concluding argument here is that educating students systematically about AI will need to take into consideration the differences between student populations – and, by inference, individuals. This requires a differentiated pedagogical approach, acknowledging various cultural, contextual and deeply personal differences between students – which is currently a weak point of most skills frameworks, as they do not shed light on their own "operationalisation" and operability. This is only aggravating the already slow adoption of such frameworks by teachers, the reasons of which we will explore in our next issue.

3.2. Issue two: the individually and structurally grounded reluctance of teachers to engage with AI

There is yet hardly any published research available on the AI proficiency of educators. What we found, though, are publications that support our case, why educators by and large might be adopting AI at a slower pace than many of their students, in particular if they are not working in fields related to ICT or methodologically driven by machine-learning and deep learning analytical methods. First of all, educators might be less inclined to engage with AI – at least officially - if their institution is not making this a priority. A recent UNESCO survey found that, at least within their sample, only about 10 % of the schools and universities had established institutional AI policies and/or guidelines by 2023 (UNESCO, 2023). Students are likely less concerned with a lack of institutional policy and employ tools at their own pace, but teachers might want to (or for compliance reasons need to) wait for institution or system level directives and regulations, at least as far as in classroom deployment is concerned. Mikeladze et al. (2024) see the fact that there is not yet a consensus on effective AI frameworks and the lack of effective implementation guidelines as another factor that could slow down adoption by educators. They are also raising the need for more subject-specific approaches, making 'translating' good practices to one

's own academic field easier.

Ng et al. (2023), identify a number of challenges hampering educators, among them teachers not being familiar enough with new technologies, potentially even originating in a lack of resources, i.e.no institutional software licenses/no reimbursement for their own licenses, no staff development trainings and little time to experiment on their own etc. This two-fold lack of resources (not enough personal knowledge and experience – not enough institutional trainings and infrastructure) is also emphasized by Altes et al. (2024).

Ng et al. also allude to the fear that AI could at one point substitute teachers, resulting in a reluctance to engage with AI; and last but not least underdeveloped competencies regarding how to use AI in their teaching – and teach about AI. There is an underlying lack of knowledge about AI, that connects many of these challenges. As Fleckenstein et al. (2024) found, even experienced teachers tend to underestimate the possibilities of AI (including the ability of generative AI to imitate deficient student writing) and overestimate their own ability to identify correct sources. A lack of firsthand experience and "exploration" of AI tools seems a likely explanation for this. Chan and Lee (2023) discovered some generational differences between GenZ students and their Gen X and Y teachers, suggesting that the latter are generally more cautious. This adds to the aforementioned disinclination to explore, and suggests that educators as a group will probably be as heterogeneous as the learners they tend to, with similar variables at play.

Overall, the argument here is that teachers' attitudes toward AI and their own AI related skills seem to be a key factor in either enabling speedier tackling of the AI skills gap mentioned in the introduction – or slowing it down. In combination with the previously discussed issue, the situation becomes even more complicated: Teachers with a rather varied background, little personal AI experience and in need of institutional direction are required to support an incredibly diverse body of learners in developing AI skills. We will return to this argument later but leaving it to the individual teacher to deal with the rise of AI will likely neither be in their best interest, nor the institutions'. Yet, as we want to show in the next section, institutionalisation is facing its own type of challenges.

3.3. *Issue three: the temporal gap between rapid AI developments and the construction of "curriculum-ready" knowledge*

Higher education institutions play a big role in knowledge production – but as far as their educational operations are concerned, knowledge and knowledge transfer to a large population is usually tackled via the concept of curricula and curricular elements, such as learning objectives and outcomes as drivers; modules and courses as building blocks; textbooks and learning materials as means; and last but not least suitable assessment forms. Curricula enable higher education institutions to reliably and effectively support learners in acquiring comparable qualification profiles – and even if we abandon the idea of curricula as a storage of "sacred" knowledge, they still offer boundaries between specialized and general knowledge and subjective experiences, i.e. everyday knowledge and the level of scientificity that characterize academic institutions (Young, 2013). Following Bernstein (2000) and his concept of *pedagogical device,* what and how students are educated about, is decided in the interplay of three different hierarchical fields: knowledge production, recontextualisation (i.e. the transformation of knowledge into curricula) and reproduction (i.e. the actual act of teaching). As Annala (2023) demonstrates, even the type of higher education institution has a profound impact on what knowledge is considered to be *legitimate* enough to be taught. It stands to reason that this becomes an even bigger issue across disciplinary and national cultures.

Temporalities, most notably duration and speed, pose a big challenge too, in the current situation. Experience suggests that convergence of skills frameworks take time (as can be seen in the example of the so called 21st century skills, cf. Kennedy and Sundberg, 2020) – and so does their translation into curricula. And without some kind of "formalization", teachers appear to be reluctant to delve into the issue on their own, at least on a broader scale. Balabdoui et al. (2024) report, for example, how only a minority of students at their institution – one of the leading Technical Universities in continental Europe – reported having experienced AI of any kind in a teaching situation in a recent survey. This might partly be due to doubts regarding the ability of GenAI to provide high quality content, in connection with concerns about plagiarism and cheating (Chan and Lee, 2023). And last but not least, the translation into curricula is severely impeded by the incredible speed with which new AI applications are launched. The "platform logic" of ChatGPT enables multiple plug ins and educators can hardly keep up with the increase, let alone analysing benefits and shortcomings of individual applications. Even comprehensive platforms, such as tilburg.ai, an initiative by the University of Tilburg (Working group "AI in education" at Tilburg University, 2024) to promote the responsible use of AI in education, cover but a fraction of available tools.

Our key argument here is, that in light of the speed that characterizes current developments in (educational) technology and the fact the knowledge codification (O'Meara and Kelliher, 2021) is still ongoing, it remains doubtful that educators and learners will be systematically provided with the necessary structures and means to tackle AI in the nearer future. We do not doubt that efforts for consolidation and convergence are to some degree necessary and even already on their way – but for the immediate future we see a risk that the gap between the demand for AI literacy and the provision of AI literate graduates will continue to grow, requiring quick win strategies that at the same time help pave the way for more holistic approaches.

4. Ways forward: low threshold suggestions

As we have tried to show, implementing a full-scale AI literacy framework faces several challenges, partly from within the frameworks and partly from the context of higher education – and will thus require time – as will the drafting of institutional policies or gaining more detailed insights into students' level of AI skills. Being aware of these delaying factors but at the same time acknowledging the need for speed in addressing the growing skills gap, we propose a series of measures that can be implemented in a shorter term, a kind of low threshold actions at course level or by individual teachers, to address the issues raised.

Our suggestions are derived from the nature of the structural challenges outlined in the previous section and are based on a pedagogical approach to how these challenges might be addressed. The last challenge determines the format of solutions we propose: given the rather slow development of curricula in higher education on the one hand, and the rapid development of AI on the other, possible solutions must have the specific characteristics. They must be quickly available and adaptable, feasible on the initiative of individual teachers, and respect the diversity of the learners and the diverse experiences with AI that they bring to the courses. At the same time, our call for immediate, low threshold solutions is not meant to be a singular initiative to deal with AI in higher education. AI-proofing our curricula will be an extensive and time-consuming endeavour, given the enormous changes that will be required to achieve this (e.g. adapting course designs, assessment formats, delivery formats or learning outcomes).

In accordance with Farrelly and Baker's suggestion that "even relatively small initiatives can have a meaningful impact" (2023), the proposed approach involves the implementation of multiple (interconnected) initiatives to generate a collective effect – and pave the way for more concerted curricular efforts in the longer run. We see the highest potential in bringing students on board as early as possible, via principles such as peer learning and co-creation; and in prioritising micro-designs that can be adapted in various courses and academic programs irrespective of their core discipline.

With regard to project management education, the approaches we propose, specifically co-creation and peer learning, have the advantage

that they promote transferable skills that are highly relevant for the complexity of managing projects, such as communication skills, teamwork and cooperation, problem solving, developing others and flexibility (Moradi et al., 2020).

4.1. Learning from each other: peer (assisted) learning

Since GenAI has transitioned from a niche product to a widely accessible technology, it is useable by anyone even without specialist skills (other than natural language). Hence many students have already gained substantial experience and skills in genAI use by experimenting with it and following its development (Chan and Hu, 2023; Kelly et al., 2023a; Leiker et al., 2023; Strzelecki, 2024). As we outlined in section 3.1, there are still large differences between (groups of) learners, and with the multitude of tools available, it is very likely that students exchange experiences and scenarios of use or their preferred tools. This constitutes an example of informal peer learning.

Peer learning (or pear assisted learning; PAL) is characterized by the fact that the people who learn from each other are in a similar situation and context. "PAL is the acquisition of knowledge and skill through active helping and supporting among status equals or matched companions. PAL is people from similar social groupings, who are not professional teachers, helping each other to learn and by so doing, learning themselves" (Topping and Ehly, 1998). Peer learning occurs in everyday life and in informal contexts when students ask their peers about course content, but it can also be translated into a more formal use within courses and facilitated by a teacher or the institution. Importantly, peer learning is characterized by its reciprocity and the fact that it should be mutually beneficial (Boud, 2014).

Peer learning has several advantages that are also specifically relevant to the context of AI skills: In peer learning, the people we learn from have encountered the same challenges in the same or a similar context. It is hence very likely that an exchange will help develop skills that are relevant to the specific situation the students or teachers find themselves in. With regard to AI literacy, ideally, students learn from their peers' experiences with AI, which can enable more proficient students to refine their approaches through critical discussion and help less skilled students improve. For the latter, it might also present less of a challenge to pose very basic questions to peers rather than to teachers. With regard to the acquisition of digital skills in the project management context, Guinan et al. (2019) found that technical ability is less relevant than the openness to learn from peers. This is also a strong argument for the integration of peer learning approaches in higher education in order to foster this openness vis-à-vis the ideas and input from colleagues and prepare students for future work contexts.

Student peer learning groups exist at a variety of universities, for example "Harvard student leaders in AI" (Harvard University, 2024) the „University of Edinburgh AI Society" (University of Edinburgh, 2024) or the UCL „Artificial Intelligence Society" (University College London, 2024) and foster the (interdisciplinary) exchange on AI. While the exchange in these peer learning groups happens outside of classes, there are also meaningful ways of integrating peer learning activities in the more formal setting of a class, for example when students work on real-life problems in teams. Collaborative, project-based learning has been a staple of AI-related pedagogical approaches for decades – yet with a focus on ICT infused projects (Ng et al., 2023).

It is evident that a straightforward solution to the second issue that we have raised in section 3.2 is to provide training for teachers. Nevertheless, it should be noted that not all teachers are inclined to attend these workshops, and this approach is often more efficient to motivate those who are already interested in AI. Consequently, facilitating peer learning among teachers could be a viable strategy, potentially in the (time-efficient) form of an online good practice hub or similar solutions that share best practices. An example of such a knowledge hub is provided by the Teach & Learn Section within the MIT AI Hub, where general knowledge on AI, practical tips for teaching with AI and also discipline-specific AI use cases are shared (MIT Sloan School of Management).

Mills et al. (2023) propose open educational practices as a possibility to move forward. Open educational practices include open educational resources, but also open pedagogies and open sharing of teaching practices. Mills et al. also provide concrete examples, such as "sharing rough work, crowd-sourcing curation, building on others' experiments, collaborating with students (also known as open pedagogy), and planning for continuous revision and reflection" (Mills et al., 2023, 18).

4.2. Tackling AI in partnership: Co-creation

Another approach that allows teachers and students to learn from each other and build something together is the concept of co-creation. As we outlined in section 3.2, for various reasons teaching staff are less likely having experimented with genAI and could benefit considerably from students' digital fluency. This dynamic makes AI skill development an ideal case for co-creation. "Co-creation of learning and teaching occurs when staff and students work collaboratively with one another to create components of curricula and/or pedagogical approaches." (Bovill et al., 2016). It is hence conceptualized as a means of making students more active participants in the learning process (Bovill et al., 2016) and allows staff and students to bring in their specific expertise and learn from each other. An important pilar in co-creation is that hierarchy is reduced throughout the process and decision-making is perceived a shared responsibility (Lubicz-Nawrocka and Bovill, 2023).

Co-creation can be applied at different levels: Bovill and Woolmer distinguish between two forms, co-creation of the curriculum, which usually happens before the program or course starts and co-creation in the curriculum, which happens during the course. Co-creating the curriculum is an interesting approach also with regard to AI literacy, as it is documented that it can lead to significant changes in students' thinking and practices and foster a deeper metacognitive awareness of learning (Lubicz-Nawrocka and Bovill, 2023). In some cases, students even carry the "partnership mindset" they acquired when engaged in co-creation into their professional careers (Peseta et al., 2020). Both outcomes are particularly valuable with regard to the development of AI competences yet require time.

Co-creation on the course level can be initiated and implemented by individual teachers. As it is a pedagogical approach, it is not tied to specific disciplinary knowledge and we consider it low threshold in the sense that it can easily be included in already existing learning activities, supplement or replace them without the prerequisite of a fully implemented AI skills framework.

Mohamed et al. (2025), from their cross-national findings, strongly advocate for gaining students' trust for AI in teaching, by bringing in their perspective and giving voice to their concerns, not least about ethics. Co-creation also increases student motivation to engage with AI tools and students tend to take more responsibility for their own learning when invited to contribute—a particularly beneficial outcome in the context of AI skills, where self-directed learning fosters self-control, adaptability and critical thinking. These qualities are essential for navigating the rapid evolution of AI technologies and applying them effectively in diverse real-world scenarios. And skills like self-control, critical-thinking and planning skills, which are fostered when students are encouraged to take responsibility for their own learning, are clearly also essential for project managers (Moradi et al., 2020).

In the case of AI skill development, co-creation positions educators as facilitators who can leverage student expertise in digital fluency while providing structure, critical reflection, and disciplinary insights to the learning process. This strong element of role-reversal or at least change of roles, inherent in co-creation might be a challenge for teachers: educating others in a field where at least some of those being educated hold an experiential advantage can be daunting – but also offers opportunities for teachers. Apart from making use of the digital fluency of

students this approach is also a tool to help educators to better understand their students' current knowledge and skills and identify areas where additional support might be needed. Beyond acting as facilitators, teaching staff play a crucial role in bringing disciplinary expertise, supporting students in fact-checking processes, advocating for reflective use of AI tools, and ensuring that intended learning outcomes remain achievable even with the general availability of generative AI tools.

With regard to AI skills development, there are examples where students are involved in the creation of study aids for their peers, for example the (publicly accessible) CANVAS module that was created by staff and students at the University of Sydney (University of Sydney, 2024). Another example Is the AI Co-Creators projects at University College London, where staff and students collaborate to design AI projects funded by the University.

4.3. Starting small via micro designs

Making students genuinely AI literate is undoubtedly a long-term endeavour, requiring comprehensive curricular integration and alignment with institutional and program goals. However, there is evidence that even micro implementations can have an impact and constitute the first step on the way to AI literacy. Balabdaoui et al. (2024) found that students who learn *about* AI, show a high level of AI proficiency. Sullivan et al. (2024) demonstrated that even a single workshop can enhance awareness and boost confidence in AI use among both students and staff, reinforcing the value of focused, short-term interventions. Similar results were reported by Kong et al. (2021) who evaluated a 7-h course on AI literacy and found that student improved their AI literacy as well as Korte et al. (2024) who discovered that students who participated in 5 h of online lectures on AI literacy gained greater confidence in applying it in their everyday lives.

Such micro-designs can address different levels, taking the form of pedagogical interventions and learning activities that can be embedded across various courses and disciplines. Adaptable to diverse academic contexts, the learning activities should aim at fostering immediate engagement with AI concepts and tools and allow students to experiment with AI and build skills (i.e. what Balabdaoui et al., 2024 call "learning *with* AI"). Examples for these micro-designs include integrating deliberate AI-based assignments, encouraging students to reflect on their experiences with AI tools or analyzing and critiquing AI-generated outputs collaboratively, which help students to understand how to use AI. From a pedagogical perspective, an emphasis on hands-on, experiential learning is crucial for the development of AI skills (Southworth et al., 2023). Transforming learning activities according to this principle includes working on real-life problems that need AI tools/techniques, innovative learning activities like hackathons or coding competitions (Southworth et al., 2023). These pedagogical approaches are inherently flexible and adaptable across disciplines, making them suitable for immediate implementation. They also often foster creativity, critical thinking, and problem-solving.

Micro-designs can also address students directly, i.e. when self-learning modules are offered that introduce students and staff to the basics of AI ("learning *about* AI" according to Balabdaoui et al., 2024). Such modules are particularly valuable as they provide scalable and self-paced learning opportunities. Developed by the institutions themselves, they also take the context into consideration, which makes the modules more relevant to staff and students. Many institutions worldwide have successfully implemented such modules, for example University College Cork with their Toolkit for the Ethical use of genAI for staff (University College Cork, 2024), MIT AI Hub "Getting Started with AI-Enhanced Teaching: A Practical Guide for Instructors" (MIT Sloan School of Management, 2023) or Tilburg University's e-module for students (Tilburg University, 2024) providing learners and faculty with a baseline understanding that can be built upon through discipline-specific applications and collaborative activities. Regarding the content, the focus in these modules is mostly on the topics that are particularly relevant for developing a basic understanding of how AI works, i.e. knowledge on large language models and AI tools as well as guidelines for an ethical use of the tools.

Last but not least, micro-designs can also serve teaching staff development, e.g. in the case of good practice hubs and exchange formats, or self-learning materials. Incentivizing the experimentation with AI tools as for example by issuing awards (e.g. AI awards at the University of Florida; University of FLorida, 2024), by providing project money for AI-related implementations in courses (e.g. AI CO-Creators Projects at UCL (University College London, 2023); are also steps to increasing the educators' AI literacy.

As the pace of AI development accelerates and the skill level among students and staff rises, the importance of integrating AI literacy into the curricula will grow. However, even at this stage, embedding micro-design elements can yield significant benefits. These interventions serve as the building blocks for a more comprehensive and contextualized approach to AI literacy.

5. Conclusion

Over decades, if not centuries, higher education institutions have developed sophisticated mechanisms and processes to enable learners to develop viable skills to help them navigate the complexities of their working lives and society at large. As we have tried to show, the rapid emergence of (generative) artificial intelligence into the public consciousness, as part of the broader meta-trend of digitalising virtually every aspect of our work and everyday lives, is challenging these mechanisms.

The need for AI-literate graduates is no longer just an issue for the ICT sector or part of "industry 4.0". Calls to equip graduates across disciplines and types of education with a basic understanding of the potential and the shortcomings of artificial intelligence, the ability to work with AI effectively and ethically and to use and develop AI reflectively are echoed across different economic and societal sectors around the world. Project management is also affected by the need for digitalization (Marnewick and Marnewick, 2021) and the need for the development of skills that will help future project managers to navigate the rapidly changing and evolving environment they will have to deal with. The call for the development of transferable skills such as flexibility, creativity or critical thinking has been voiced earlier (e.g. Pant and Baroudi, 2008) but becomes particularly relevant in the light of the current challenges posed by AI.

As our analysis has shown, a number of factors are currently hindering or at least slowing down the systematic response of higher education institutions to the call for the integration of AI skills: The frameworks that will ultimately be needed for concerted efforts are not yet mature enough, nor is there a consensus on what to prioritise – and, even more importantly, how to operationalise existing frameworks. In addition, learners are more diverse than ever, with a wide range of biographical, personal, socio-economic and cultural factors influencing their AI readiness. The same is true for teachers, who additionally have to cope with a lack of institutional direction in the form of educational policies that provide orientation, not least on ethical questions. And finally, the dissemination of knowledge validated by the scientific communities, which is a necessary prerequisite for the creation of curricula and pedagogical strategies to support education, is challenged by the incredible speed of technological development.

Of course, this is not an argument against immediate action: Swift solutions are needed now more than ever. At a metalevel, this could mean prioritising pragmatic approaches to skills development in favour of comprehensive competence frameworks. There are many lessons to be learnt from experience with the Framework for 21st Century Skills and similar frameworks, not least because some of the transversal competences (critical thinking, innovation, creativity and ethics) overlap. There is a substantial body of literature on these topics (see Abrami et al., 2015; Andrews, 2015; Ennis, 2015; Geisinger, 2016), suggesting

that there is room to learn from the discourses when it comes to teaching these competences in the field of AI. Two key insights are the challenge of assessing learning outcomes, and the question how transferable competences can be operationalized in a pedagogical sense, i.e. how they can be brought to life in teaching and successfully be taught and transferred to different disciplines.

We argue that pedagogical strategies – co-creation, peer-assisted learning, and micro-designs – can be a solution to tackle the challenges posed by AI. Additionally, these approaches are particularly well suited to the context of project management education, where students must navigate complexity, collaboration, and constant change. Peer assisted learning and co-creation are pedagogical approaches that reinforce a certain set of transferable skills, such as analytical thinking, creativity, problem solving, teamwork and cooperation, proactivity, self-control, emotional resilience, and interpersonal skills such as interpersonal understanding, cultural skills and team management, which are equally relevant for project managers (Moradi et al., 2020). Teaching AI skills and responsible use of tools through these pedagogies not only enhances AI literacy but also models the adaptive, agile and collaborative ethos that defines effective project management in the 21st century.

In awareness of the fact, that these efforts will take time, and can rarely be attempted by individuals or small groups of actors, we believe that investing in low threshold approaches, such as micro design elements and pedagogical principles like peer-assisted learning and co-creation, will help to address the most burning needs without overwhelming teachers and learners alike. While the suggested solutions have a clear benefit in providing quick and easily contextualizable approaches, there is also an inherent limitation: Individual approaches are often futile and rely on personal motivation and the initiative of individuals. So, in order to make these small-scale innovations a stepping stone for systematic and strategic processes to address the structural challenges, they need to be institutionalised and – ultimately – be "codified" in curricula that integrate AI skills at different levels and are agile enough to adapt to the ever-occurring changes.

CRediT authorship contribution statement

Oliver Vettori: Writing – review & editing, Writing – original draft, Conceptualization. **Johanna Warm:** Writing – review & editing, Writing – original draft, Conceptualization.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used DeepL and ChatGPT in order to improve language and readability. After using this tools/services, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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